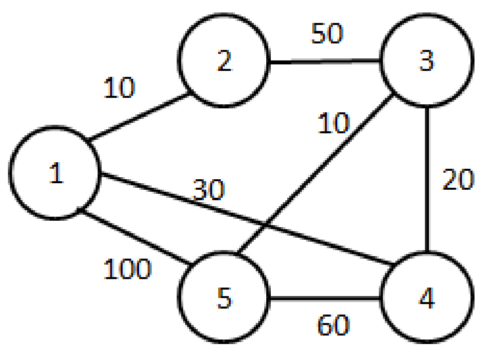
Internal Assessment Question Paper – 2

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Artifiial Intelligence and Data Science

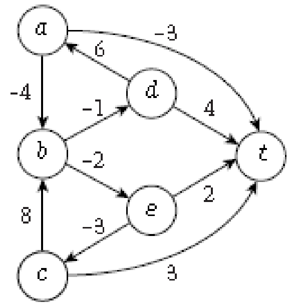
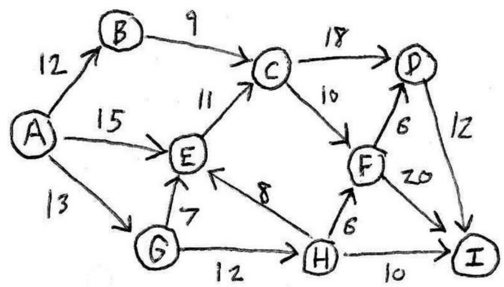
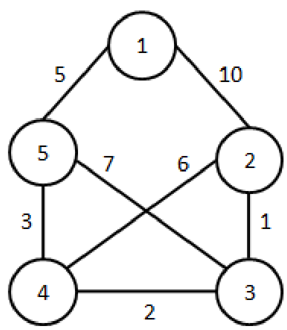
Programme: B.E **Term:** June-Sept 2023 **Max Marks:** 30
CIE: Test 2 **Subject:** Design and Analysis of Algorithms **Subject Code:** AD43/AI43
Crédits: 3:0:0:0
Sem: IV **Date: 26/07/2023** **Time: 2:00 to 3:00** **Portions for Test:** (L21-L42)

Instructions to Candidates:

1. **Question 1 is Compulsory.** Answer any one question from 2 and 3.
2. Each Question carries 15M. **Write figures wherever necessary.**
3. Mobiles/Smart Gadgets are restricted.

<table>
<tr><th>Sl#</th><th>Question</th><th>Marks</th><th>Bloom's Level</th><th>CO Mapping</th></tr>
<tr><td>1</td><td>a. Construct the Huffman tree and the prefix code for the given set of alphabets and their frequencies. S={a,b,c,d,e}
f_a=0.40, f_b=0.1, f_c=0.25, f_d=0.2, f_e=0.05</td><td>04</td><td>Apply</td><td>CO3</td></tr>
<tr><td></td><td>b.
i. Identify the shortest path for the following graph from source '1' to all other nodes. (5M)


ii. Construct a proof for "The greedy algorithm returns an optimal set A" for a given set of intervals with their start time (si) and finish time (fi). (4M)</td><td>09</td><td>Apply</td><td>CO3</td></tr>
<tr><td></td><td>c. Define P and NP.</td><td>02</td><td>Remember</td><td>CO2</td></tr>
<tr><td>2</td><td>Identify the items along with maximum profit that belong to the knapsack with maximum capacity 6 for the following.
<table><tr><th>Item</th><th>Weight</th><th>Value</th></tr><tr><td>1</td><td>2</td><td>10</td></tr><tr><td>2</td><td>3</td><td>20</td></tr><tr><td>3</td><td>4</td><td>10</td></tr><tr><td>4</td><td>5</td><td>30</td></tr></table></td><td>05</td><td>Apply</td><td>CO4</td></tr>
<tr><td></td><td>b. Identify the shortest path for the given graph from each node to 't' using Bellman ford algorithm.</td><td>06</td><td>Apply</td><td>CO4</td></tr>
</table>

				
	c. Design an algorithm to determine the lateness for a given set of intervals with their time duration(t_i) and deadlines(d_i).	**04**	Apply	CO3
3	a. Identify the maximum flow F_M for the following graph from "A-> I" using Ford-Fulkerson algorithm. 	**06**	Apply	CO4
	b. Illustrate Optimal Caching Technique	**04**	Apply	CO3
	c. Construct the minimum spanning tree for the following graph using Prim's algorithm. 	**05**	Apply	CO3

Course Outcomes meant to be assessed by the IA Test 2:

CO3: Illustrate the design techniques for divide and conquer algorithms and analyze their complexity by solving recurrence relations. (PO1, PO2, PO3, PO4, PO5, PO12, PSO2)

CO4: Illustrate Greedy paradigm and Dynamic programming paradigm using representative algorithms. (PO1, PO2, PO3, PO4, PO5, PO12, PSO2)

CO5: Describe the classes P, NP, and NP-Complete and be able to prove that a certain problem is NP-Complete and examine the techniques of proof by contradiction, mathematical induction and recurrence relation, and apply them to prove the correctness of the running time of algorithms. (PO1, PO2, PO3, PO4, PO5, PO12, PSO2)